REPORT ON

N-P-K FERTILIZER UTILIZATION RECOMMENDATION SYSTEM

AT

PANAPPS INTERNATIONAL PVT. LTD.

Submitted to

Cochin University of Science and Technology

In partial requirement for the award of the degree of

BACHELOR OF VOCATION

IN

BUSINESS PROCESS AND DATA ANALYTICS

Submitted By,

ANJANA BABU

(Reg no: 25123011)

Under the guidance of

Ms. Joanna Paul

Assistant Professor, DDUKK

DDU KAUSHAL KENDRA

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Kochi – 682022

May 2026

CERTIFICATE FROM THE ORGANIZATION

PanApps

CIN: U72200KL1999PTC018425

HRD/PAS/2026/04/060 20/04/2026

TO WHOMSOEVER IT MAY CONCERN

This is to certify that Ms.Anjana Babu has successfully completed her internship in the R&D division of PanApps International Pvt Ltd for a period starting from 15th, December 2025 to 17th, April 2026.

During the above period, she has been assigned the following tasks on:

- Predictive, Classification and Clustering problems.
- Data preprocessing, and model evaluation.
- Data Visualization using PowerBI.
- Semantic Search powered AI Application.

During her internship we found that she was consistent and self motivated to learn new skills, and diligent in her assigned duties and responsibilities.

We wish her all success in her future endeavors.

For PanApps International Pvt. Ltd.
[W.O.S of PanApps Inc.VA.USA]

Salij TS
Associate Vice President – Human Resources

Confidential

PanApps International Pvt. Ltd. [Wholly Owned Subsidiary of PanApps Inc,VA,USA]
Door No.XIV/693,Muttom, Kochi-683106, Kerala Email: hr@panapps.co , www.panapps.co

CERTIFICATE FROM THE DEPARTMENT

DEEN DAYAL UPADHYAY KAUSHAL KENDRA

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

KOCHI- 682022

This is to certify that this report on NPK Fertilizer utilization recommendation system a**t** PanApps International Pvt. Ltd. done as part of internship submitted to Cochin University of Science and Technology for the partial fulfilment of the award of the Degree of Bachelor of Vocation in Business Process and Data Analytics under the Faculty of Social Sciences, is the record of bona fide work done by Ms. Anjana Babu, Reg no: 25123011 under my supervision and guidance.

Ms. Joanna Paul
Supervising Guide
Asst. Professor

Prof. (Dr.) Santhosh Kumar S.
Director, DDUKK

Kochi- 682022

Date:

DECLARATION

I , Anjana Babu, sixth semester B.Voc in Business Process and Data Analytics student of DDU Kaushal Kendra at Cochin University of Science and Technology hereby declare that this report on N-P-K FERTILIZER UTILIZATION RECOMMENDATION SYSTEM at PanApps International Pvt Ltd under the guidance of Ms. Joanna Paul., Asst. Professor, DDU Kaushal Kendra in partial fulfilment of the requirement of the degree of Bachelor of Vocation in Business Process and Data Analytics is a bonafide study I have done in the organisation. I also declare that this report has not been previously formed on the basis of award of any Degree, Diploma, Fellowship or any other similar title of this or any other university or institution.

Anjana Babu

Reg no:

25123011

Place :

Date :

ACKNOWLEDGEMENT

I would like to thank God Almighty, for giving me the strength, patience, and ability to complete the study and make this report on time.

I am immensely grateful to Ms. Divya R V and Ms. Sreekumari , my mentors at PanApps, for their guidance, encouragement, and continuing support which they rendered throughout the study and was instrumental in shaping this report.

My sincere gratitude goes to my guide, Ms. Joanna Paul , Asst. Professor, DDU KAUSHAL Kendra, CUSAT for being a source of encouragement throughout this study. Her valuable suggestions have enabled me to complete my reports to perfection.

I also thank the department, especially our director Prof. (Dr.) Santhosh Kumar S , and all other faculty for giving me such wonderful exposure during my degree.

ABSTRACT

This internship project, carried out as part of the B.Voc curriculum in Business Process and Data Analytics, focused on developing a data-driven fertilizer recommendation system using machine learning techniques.

The study involved analyzing a structured dataset representing real-world agricultural conditions, including features such as soil nutrient levels, crop type, irrigation methods, and climatic factors. The primary goal was to build a predictive model capable of recommending appropriate quantities of soil nutrients for efficient crop production.

The project followed the CRISP-DM methodology, covering key stages from data understanding to model evaluation. Multiple regression models were trained and evaluated, among which Gradient Boosting demonstrated the best performance based on evaluation metrics.

Additionally, a user-friendly interface was developed using Gradio, allowing users to input agricultural parameters and receive fertilizer recommendations in both nutrient form and commonly used fertilizers.

The outcomes of this project highlight the potential of machine learning in improving agricultural practices by optimizing fertilizer usage, reducing wastage, and supporting informed decision-making. This system can serve as a practical tool for enhancing productivity and promoting sustainable farming.

TABLE OF CONTENTS

SL NO	CONTENT	PAGE NO
	CERTIFICATE FROM THE ORGANIZATION	ii
	CERTIFICATE FROM THE INSTITUITION	iii
	DECLARATION	iv
	ACKNOWLEDGEMENT	v
	ABSTRACT	vi
1	**ABOUT THE STUDY**	**7**
1.1	ABOUT THE STUDY	8
1.2	PROBLEM STATEMENT	8
1.3	OBJECTIVES OF THE STUDY	8 - 9
1.4	SCOPE OF THE STUDY	9
1.5	METHODOLOGY OF THE STUDY	9 - 14
1.5.1	CRISP-DM METHODOLOGY	10 - 11
1.5.2	SIX KEY PHASES OF CRISP-DM METHODOLOGY	11 - 14
1.6	LIMITATIONS OF THE STUDY	14
2	**BUSINESS UNDERSTANDING**	**15**
2.1	INTRODUCTION	16
2.2	COMPANY OVERVIEW	16

2.3	CORE DIGITAL SERVICES	16 - 17
2.4	INDUSTRY SPECIFIC PRODUCT SOLUTIONS	17 - 18
3	**DATA UNDERSTANDING**	**19**
3.1	DATA GATHERING	20
3.2	DATA UNDERSTANDING	21 - 22
3.2.1	DESCRIPTION OF DATASET ATTRIBUTES	22 - 23
3.3	LIBRARIES USED	24
3.4	EXPLORATORY DATA ANALYSIS	25 - 28
3.4.1	DISTRIBUTION OF CROP	25
3.4.2	DISTRIBUTION OF SEASON	26
3.4.3	DISTRIBUTION OF SOILTYPE	26 - 27
3.4.4	CORRELATIONN HEATMAP OF NUMERICAL FEATURES	27 - 28
4	**DATA PREPARATION**	**29**
4.1	DATA PREPROCESSING	30
4.1.1	HANDLING MISSING VALUES	30
4.1.2	ENCODING CATEGORICAL VALUES	30 - 31
4.1.3	FEATURE SELECTION & DATA SPLITTING	31
4.1.4	FEATURE SCALING	31 - 32
5	**MODEL BUILDING**	**33**
5.1	MODEL BUILDING	34
5.1.1	KEY BENEFITS OF DATA MODELING IN THIS PROJECT	34

5.2	LINEAR REGRESSION	36 - 37
5.3	RANDOM FOREST REGRESSOR	37
5.4	GRADIENT BOOSTING REGRESSOR	38
6	**MODEL EVALUATION**	**39**
6.1	MODEL EVALUATION	40
6.1.1	IMPORTANCE OF MODEL EVALUATION	40 - 41
6.2	PERFORMANCE MEASUREMENT METRICS	41 - 42
6.3	MODEL PERFORMANCE ANALYSIS	42 - 45
6.4	FEATURE IMPORTANCE	45 - 46
6.5	ACTUAL VS PREDICTED	47 - 48
7	**DEPLOYMENT**	**49**
7.1	TOOLS AND TECHNOLOGIES USED	50
7.2	APPLICATION OF ARCHITECTURE	51 - 52
7.3	USER INTERFACE DESCRIPTION	52
7.4	LIMITATIONS OF PROTOTYPE	52 - 53
	CONCLUSION	54
	FUTURE SCOPE	55
	BIBILIOGRAPHY	56
	ANNEXURE	57 - 59

LIST OF FIGURE

FIG NO	FIGURE	PAGE NO
1.1	PHASES OF CRISP-DM METHODOLOGY	16
3.1	TABLE OF DESCRIPTION OF DATASET ATTRIBUTES	28
3.2	DISTRIBUTION OF CROP	30
3.3	DISTRIBUTION OF SEASON	31
3.4	DISTRIBUTION OF SOIL TYPE	31
3.5	CORRELATION HEATMAP	32
4.1	HANDLING MISSING VALUES	35
4.2	ENCODING CATEGORICAL VARIABLES	36
4.3	TRAIN – TEST SPLIT	36
4.4	FEATURE SCALING	37
5.1	MODELS USED	40
5.2	DEFINING MODELS	40
5.3	CALCULATING R^2 SCORES	41
5.4	CALCULATING MAE & RMSE	41
6.1	R^2 SCORE RESULTS	47
6.2	MAE & RMSE RESULTS	47
6.3	BARPLOT COMPARISON OF INDIVIDUAL R^2 SCORES	49
6.4	BARPLOT COMPARISON OF AVERAGE R^2 SCORES	49
6.5	BARPLOT COMPARISON OF MAE	50
6.6	FEATURE IMPORTANCE	51
6.7	ACTUAL VS PREDICTED	52
7.1	USER INTERFACE OF GRADIO APP	56

LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
DAP	Diammonium Phosphate
CRISP-DM	Cross-Industry Standard Process for Data Mining
ETL	Extract Transform Load
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
GCD	Greatest Common Divisor

CHAPTER 1

ABOUT THE STUDY

1.1 ABOUT THE STUDY

Agriculture remains the backbone of the Indian economy, supporting the livelihoods of millions of farmers across diverse agro-climatic zones. Among the many inputs that determine crop productivity, fertilizer application plays a decisive role. Nitrogen (N), Phosphorus (P), and Potassium (K) are the three primary macronutrients essential for plant growth, and their balanced application is critical to achieving optimal yields while preserving long-term soil health.

Despite their importance, fertilizer usage in Indian agriculture is largely driven by habit, generalized recommendations, or dealer advice rather than scientific assessment of actual soil conditions and crop requirements. This blanket approach to fertilizer application has led to widespread nutrient imbalances — with some soils experiencing excessive nitrogen loading while remaining deficient in phosphorus or potassium. The consequences are far-reaching, including declining soil fertility, reduced crop yields, increased input costs for farmers, and environmental concerns such as groundwater contamination and greenhouse gas emissions from over-application of nitrogenous fertilizers.

The need for a data-driven, crop-specific, and location-aware fertilizer recommendation system has therefore become increasingly relevant in the context of modern precision agriculture. Such a system would enable farmers to move away from blanket fertilizer schedules and toward targeted nutrient management — applying the right nutrient, in the right quantity, at the right time, for the right crop

1.2 PROBLEM STATEMENT

To design and implement a machine learning-based N-P-K Fertilizer utilization Recommendation System that suggests optimal fertilizer quantities based on soil parameters (N, P, K, pH), crop type, and location data, aiming to maximize yield while minimizing environmental impact.

1.3 OBJECTIVES OF THE STUDY

1. Analyze soil health metadata and environmental parameters (N, P, K, pH, crop type, and location) to establish a baseline for nutrient deficiency patterns.

2. Identify correlations between soil conditions and crop requirements through Exploratory Data Analysis (EDA) to optimize the precision of fertilizer recommendations.
3. Address the business and data needs of sustainable agriculture by designing a system that maximizes crop yield while minimizing the environmental impact of over-fertilization.
4. Evaluate and implement a machine learning pipeline to predict optimal N-P-K quantities, ensuring the model's outputs meet proven confidence levels and error standards.
5. Develop a functional user interface that receives real-time user inputs and translates raw chemical requirements (N, P, K) into actionable commercial fertilizer doses like Urea and DAP.
6. Perform sensitivity analysis and model validation to ensure the system is scalable and adaptable to diverse geographic regions and soil types.

1.4 SCOPE OF THE STUDY

The study aims to provide a comprehensive framework for designing and implementing a machine learning-based N-P-K Fertilizer Recommendation System utilizing the CRISP-DM methodology. It focuses on the systematic integration of soil parameters (N, P, K, pH), crop types, and location data to generate precise nutrient recommendations.

The project is grounded in data analysis and modeling phases—including data cleaning, preprocessing, and the evaluation of predictive algorithms—to ensure high-confidence outputs. Furthermore, the scope extends to the development of a functional interface that translates technical chemical requirements into actionable commercial fertilizer quantities (e.g., Urea, DAP), providing a reportable action plan to optimize agricultural yields while minimizing environmental impact.

1.5 METHODOLGY OF THE STUDY

The project was carried out as a 80-working-day internship at PanApps under the guidance of industry mentors and internal academic guides. The methodology strictly followed the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, which is a systematic approach to data analytics project execution.

1.5.1 CRISP-DM Methodology

The CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology is a widely recognized framework used for structuring and executing data analytics and machine learning projects. It provides a systematic and organized approach that guides practitioners through the entire lifecycle of a data mining project, from understanding the problem to deploying the final solution.

CRISP-DM is widely adopted across industries because of its practicality, flexibility, and structured workflow**.** Although the framework defines a sequence of stages, it is inherently iterative, meaning that practitioners often revisit earlier phases to refine their analysis and improve results as new insights are discovered.

In this project, the CRISP-DM framework was adopted to guide the development of the fertilizer recommendation system using machine learning. The methodology helped ensure a structured approach to understanding the data, preparing it for modeling, developing predictive models, evaluating their performance, and building an interface for practical usage.

The CRISP-DM framework consists of six key phases, which were followed during the execution of this project.

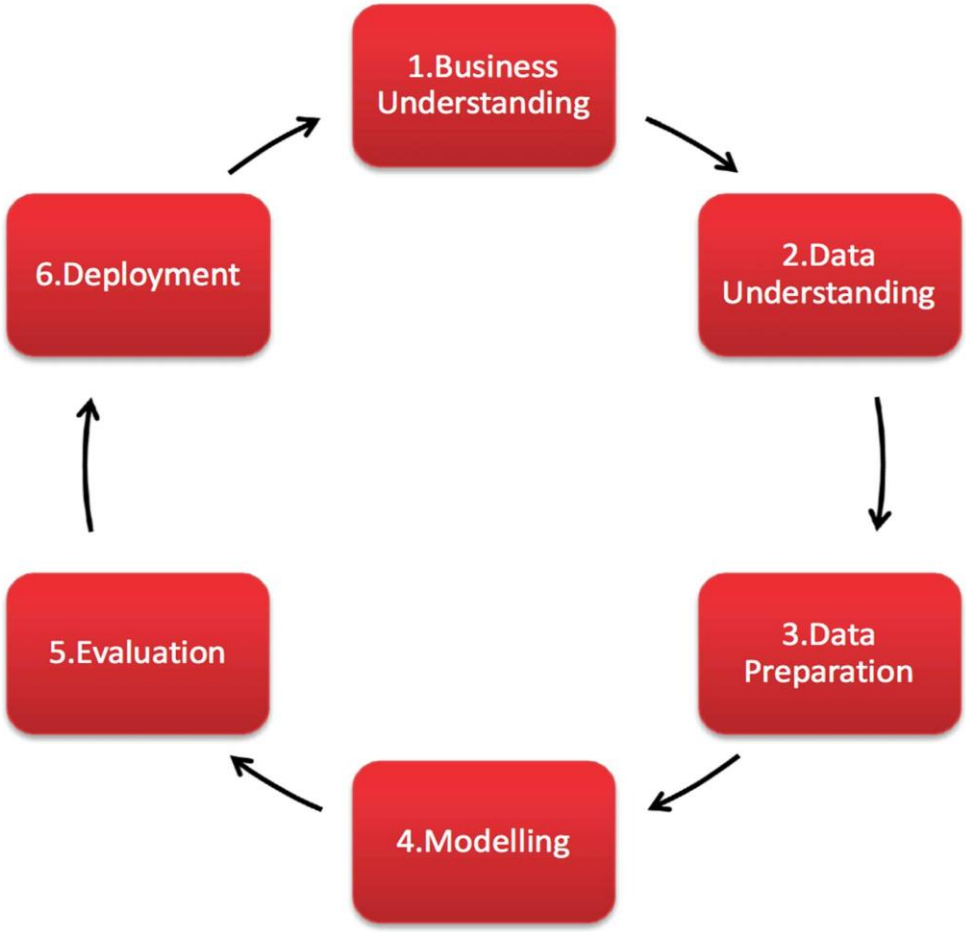

Fig 1.1 Phases of CRISP-DM methodology

1.5.2 Six Key Phases of the CRISP-DM Methodology

1. Business Understanding

The first phase focuses on understanding the problem domain and project objectives. In the context of this study, the goal was to develop a system that can recommend optimal fertilizer quantities based on soil and environmental parameters.

Farmers often face challenges in determining the correct amount of fertilizer required for crops. Excessive or insufficient fertilizer application can negatively affect crop productivity and soil health. Therefore, the objective of this project was to develop a machine learning model capable of predicting appropriate fertilizer requirements in terms of Nitrogen (N), Phosphorus (P), and Potassium (K).

This phase involved defining the project objectives, identifying the expected outputs, and determining how machine learning could be used to support fertilizer decision-making.

2. Data Understanding

The data understanding phase involves collecting and exploring the dataset in order to understand its structure, attributes, and overall quality.

During this stage, the dataset used for fertilizer prediction was examined to identify the available features and their characteristics. The following activities were performed:

- Inspecting the dataset structure and size
- Understanding the meaning of each attribute
- Checking for missing values or inconsistencies
- Analyzing the distribution of variables
- Identifying relationships between input features and fertilizer requirements

Initial exploratory analysis helped gain insights into the dataset and provided a better understanding of how different variables influence fertilizer recommendations.

3. Data Preparation

The data preparation phase focuses on transforming raw data into a format suitable for machine learning algorithms. This stage is often the most time-consuming step in a data mining project.

In this project, several preprocessing techniques were applied to prepare the dataset for modeling. These included:

- Cleaning the dataset by handling missing or inconsistent values
- Encoding categorical variables using One-Hot Encoding
- Selecting relevant features for modeling
- Preparing the dataset for machine learning algorithms

These preprocessing steps ensured that the dataset was properly structured and compatible with the machine learning models used in the study.

4. Modeling

The modeling phase involves selecting appropriate machine learning algorithms and training models using the prepared dataset.

In this project, multiple regression-based machine learning models were implemented to predict fertilizer quantities. Since the system predicts three outputs simultaneously (Nitrogen, Phosphorus, and Potassium)**,** a multi-output regression approach was used.

Several models were trained and tested, including:

- Linear Regression
- Random Forest Regression
- Gradient Boosting Regression

The dataset was divided into training and testing sets to evaluate how well the models generalize to unseen data. Each model was trained using the training dataset and evaluated using the testing dataset.

5. Evaluation

The evaluation phase focuses on assessing the performance of the trained models to determine their effectiveness in solving the problem.

In this study, the performance of the models was evaluated using the following metrics:

- R^2 Score (Coefficient of Determination) - measures how well the model explains the variance in the data.
- Mean Absolute Error (MAE) - measures the average difference between predicted and actual values.

After comparing the performance of all models, Gradient Boosting Regression demonstrated the best results based on these evaluation metrics. Therefore, it was selected as the final model for generating fertilizer recommendations.

6. Deployment

The final phase of the CRISP-DM methodology involves deploying the developed solution so that it can be used in practical scenarios.

In this project, an interface was developed that allows users to input selected parameters related to soil and environmental conditions. The trained machine learning model processes these inputs and predicts the optimal fertilizer requirements in terms of Nitrogen (N), Phosphorus (P), and Potassium (K).

Additionally, the predicted nutrient values are converted into commonly used fertilizers such as Urea and DAP, making the recommendations more practical and understandable for users.

The developed system demonstrates how machine learning can assist in supporting agricultural decision-making and improving fertilizer management practices**.**

1.6 LIMITATIONS OF THE STUDY

- Time restriction was the main constraint for project execution.
- The study could have been more effective with an extended duration.
- Data processing and analysis were limited by the availability and quality of the collected data.
- The complexity of data collection methods may have introduced biases or inconsistencies in the dataset.

CHAPTER 2

BUSINESS UNDERSTANDING

2.1 INTRODUCTION

PanApps, legally operating as PanApps International Private Limited, is a high-growth technology firm specializing in Cloud Application Software Services and Platform as a Service (PaaS). Founded with the mission to bridge the gap between complex enterprise requirements and agile digital solutions, PanApps has established itself as a strategic partner for organizations seeking to modernize their digital infrastructure.

The company distinguishes itself by moving beyond traditional software development to offer platform-driven frameworks. By integrating emerging technologies such as Big Data analytics, Artificial Intelligence (AI), and Machine Learning (ML), PanApps enables businesses to transition from legacy systems to scalable, cloud-native environments. Their approach focuses on creating a "connected enterprise" where data flows seamlessly across departments to drive innovation and competitive advantage.

2.2 COMPANY OVERVIEW

PanApps is dedicated to delivering efficient, secure, and highly customizable digital ecosystems. The company's operations are ISO/IEC 27001 certified, ensuring that information security management is at the core of every solution. Below is a detailed breakdown of its key services and specialized product divisions:

2.3 CORE DIGITAL SERVICES

PanApps provides the foundational expertise required to navigate the modern tech landscape:

- Application Modernization**:** Rather than simple "lift-and-shift" migrations, PanApps uses proprietary frameworks to re-architect legacy applications. This process enhances software performance, reduces technical debt, and ensures compatibility with modern cloud environments.
- DevOps & Automation**:** The company facilitates a culture of continuous integration and continuous delivery (CI/CD). By optimizing workflows and automated notifications, PanApps helps clients reduce time-to-market for their own software products.

- Digital Transformation Consulting**:** PanApps acts as a consultant and implementer, helping firms integrate digital technology into all areas of business, fundamentally changing how they operate and deliver value to customers.

2.4 INDUSTRY SPECIFIC PRODUCT SOLUTIONS

PanApps has developed a suite of specialized applications designed to solve "niche-yet-critical" business challenges:

- Business Intelligence (PanBI): This is a cornerstone of their data analytics offering. PanBI allows for the extraction, transformation, and loading (ETL) of structured, unstructured, or semi-structured data. It provides users with exception-based reporting and AI-enabled visualizations to improve the effectiveness of executive decision-making.
- Agri-Business Management (PanAgro): A highly specialized ERP-style solution for the agriculture sector. It manages complex multi-subsidiary structures, handling everything from union/labor wage settlements and payroll to multi-currency and multi-crop legal entity reporting.
- Grants Management System: A full-lifecycle platform tailored for NGOs and multilateral agencies. It tracks the journey of a grant from the initial proposal and review phase through to award distribution and outcome monitoring, ensuring total transparency and accountability.
- Knowledge Management (PanKM): This platform delivers business value by harvesting and cross-linking corporate content. It enables enterprises to capture "tribal knowledge" and stored assets, making them searchable and collaborative across the entire organization.

Sectors Served

PanApps maintains a diverse portfolio, providing mission-critical support to the following industries:

- Agribusiness & Plantations
- Banking, Finance & Insurance (BFSI)
- Non-Profit Organizations & NGOs
- Multilateral Agencies
- Retail & Hospitality

CHAPTER 3

DATA UNDERSTANDING

3.1 DATA GATHERING

The dataset used for this project was provided by my project guide at Panapps Technologies as part of the internship engagement. The dataset was prepared to represent typical agricultural conditions and soil characteristics that influence fertilizer requirements. It reflects realistic combinations of soil nutrients and environmental parameters that are commonly considered when determining appropriate fertilizer application in agricultural practices.

The data contains structured information related to soil nutrient composition and environmental factors that affect crop growth. Key attributes in the dataset include parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, rainfall, and crop type, which serve as important indicators for determining the optimal fertilizer requirements. These attributes form the basis for developing a machine learning model capable of predicting suitable fertilizer quantities.

The dataset was provided in the form of a structured tabular file, making it suitable for data analysis and machine learning workflows. The structured nature of the data allowed efficient execution of tasks such as data preprocessing, exploratory data analysis, model development, and evaluation.

The dataset shared for this project does not contain any confidential or personally identifiable information. Instead, it represents generalized agricultural characteristics designed for analytical and educational purposes. All necessary precautions were taken to ensure ethical use of the dataset in accordance with the organization's data handling practices.

Key Aspects of Data Gathering

• Source: Dataset provided by the project guide at Panapps Technologies

• Format: CSV file (.csv) containing structured tabular data

• Type of Data: Structured data - soil nutrient values and environmental parameters representing agricultural conditions

• Purpose: To support the development of a machine learning model for predicting optimal fertilizer quantities in terms of Nitrogen (N), Phosphorus (P), and Potassium (K)

This dataset forms the primary input for subsequent stages of the project, including data preprocessing, exploratory data analysis (EDA), model training, and model evaluation**.** The dataset structure and variety of parameters make it suitable for applying supervised machine learning techniques to build an effective fertilizer recommendation system.

3.2 DATA UNDERSTANDING

The data understanding phase focuses on exploring and analyzing the dataset to gain insights into its structure, attributes, and overall quality before proceeding with preprocessing and modeling.

The dataset used in this project contains 11,256 records and 27 attributes, representing various agricultural, soil, and environmental parameters that influence fertilizer requirements. Each record corresponds to information related to a specific farm and crop cultivation scenario.

The dataset includes several categories of variables such as:

- Farm and location information: Farm ID, State, and District
- Crop and cultivation details: Crop type, Season, Years of cultivation, Farm size, and Previous yield
- Soil characteristics: Soil type, soil pH, organic carbon content, electrical conductivity, and micronutrients such as sulphur and zinc
- Soil nutrient levels: Nitrogen, Phosphorus, and Potassium present in the soil
- Environmental conditions: Temperature, humidity, and rainfall
- Fertilizer recommendations: Recommended quantities of Nitrogen, Phosphorus, and Potassium
- Economic indicators: Estimated fertilizer cost per acre and total fertilizer cost

During this phase, the dataset was examined to understand the data types, attribute distributions, and potential relationships between soil parameters, environmental conditions,

and fertilizer requirements. Initial checks were also performed to identify missing values, inconsistencies, or anomalies that could affect the modeling process.

Understanding the dataset helped in identifying the input features and target variables for the machine learning models. In this project, the primary target variables are the recommended quantities of Nitrogen, Phosphorus, and Potassium, which are predicted based on the various soil, environmental, and cultivation-related attributes present in the dataset.

This exploration provided the foundation for the next stages of the project, including data preprocessing, exploratory data analysis, and machine learning model development.

3.2.1 Description of Dataset Attributes

FEATURE	DESCRIPTION
FarmID	Unique identifier assigned to each farm record in the dataset.
State	The state where the farm is located.
District	The district within the state where the farm is situated.
Crop	Type of crop cultivated on the farm.
Season	Agricultural season during which the crop is grown (e.g., Kharif, Rabi).
SoilType	Category of soil present in the farm area (e.g., clay, sandy, loamy).
IrrigationType	Method used to supply water to crops (e.g., drip irrigation, canal irrigation).
SoilNitrogen_kg_ha	Amount of nitrogen present in the soil measured in kilograms per hectare.
SoilPhosphorus_kg_ha	Amount of phosphorus present in the soil measured in kilograms per hectare.
SoilPotassium_kg_ha	Amount of potassium present in the soil measured in kilograms per hectare.
SoilPH	Measure of soil acidity or alkalinity, which influences nutrient availability.

OrganicCarbon_pct	Percentage of organic carbon present in the soil, indicating soil fertility.
Sulphur_ppm	Concentration of sulphur in the soil measured in parts per million.
Zinc_ppm	Amount of zinc micronutrient present in the soil measured in parts per million.
ElectricalConductivity_dS_m	Measurement of soil salinity, indicating the concentration of soluble salts in the soil.
Temperature_C	Average temperature of the environment in degrees Celsius.
Humidity_pct	Relative humidity level of the environment expressed as a percentage.
Rainfall_mm	Amount of rainfall received in millimeters.
FarmSize_acres	Total area of the farm measured in acres.
PreviousYield_kg_acre	Crop yield obtained in the previous cultivation cycle measured in kilograms per acre.
YearsOfCultivation	Number of years the land has been used for agricultural cultivation.
NitrogenRecommendation_kg_acre	Recommended quantity of nitrogen fertilizer required per acre.
PhosphorusRecommendation_kg_acre	Recommended quantity of phosphorus fertilizer required per acre.
PotassiumRecommendation_kg_acre	Recommended quantity of potassium fertilizer required per acre.
ExpectedYieldImprovement_pct	Estimated percentage improvement in crop yield after applying recommended fertilizers.
EstimatedFertilizerCost_INR_acre	Estimated cost of fertilizer application per acre in Indian Rupees.
EstimatedFertilizerCost_INR_total	Total estimated fertilizer cost for the entire farm area.

Fig 3.1 Table of description of dataset attributes

3.3 LIBRARIES USED

- NumPy:

  This library is used for numerical computing and array manipulation. It provides support for handling large multi-dimensional arrays and matrices, along with a collection of mathematical functions to perform efficient numerical operations.
- Pandas:

  Pandas is a powerful data manipulation library that provides data structures such as DataFrame and Series. It is used for data cleaning, transformation, and analysis, making it essential for handling structured datasets.
- Matplotlib:

  Matplotlib is a plotting library used for creating various types of visualizations. It helps in understanding data trends, distributions, and relationships during exploratory data analysis.
- Seaborn:

  Seaborn is built on top of Matplotlib and provides a higher-level interface for statistical data visualization. It simplifies the creation of complex plots and enhances visual appeal with better default styles.
- Scikit-learn:

  Scikit-learn is a machine learning library that provides a wide range of algorithms and tools for data preprocessing, model building, training, and evaluation. It is extensively used in this project for implementing regression models and performance evaluation.
- XGBoost:

  XGBoost is an optimized gradient boosting library known for its efficiency and high performance. It is used for building powerful regression models.
- Joblib:

  Joblib is used for saving and loading trained machine learning models and preprocessing components for later use.
- Gradio:

  Gradio is a Python library used to create an interactive web interface for the model. It allows users to input data and receive fertilizer recommendations in real time.

3.4 EXPLORATORY DATA ANALYSIS

3.4.1 Distribution of crop

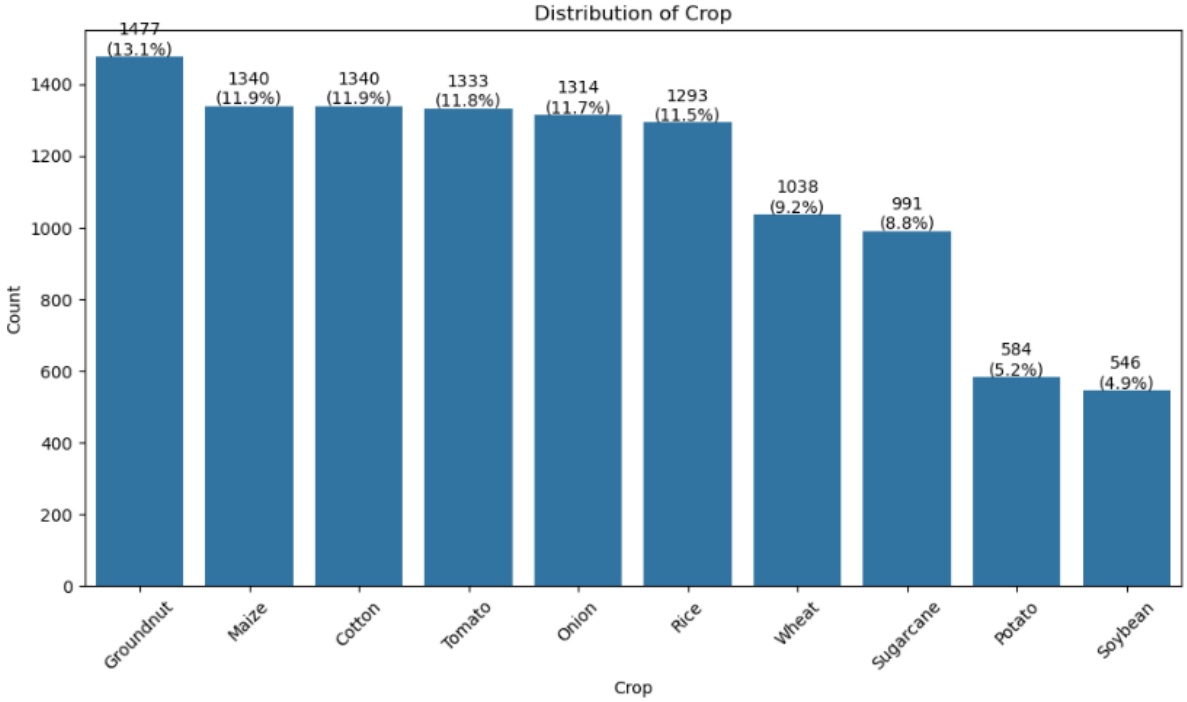

Fig 3.2 Distribution of crop

Key Insights:

As shown in the above figure 3.1, the dataset is dominated by Rice and Maize crops (45.2%), followed by commercial crops like Sugarcane and Cotton (32.8%). Other varieties such as Pulses and Oilseeds have lower representation, indicating a focus on major high-nutrient-consuming crops.

3.4.2 Distribution of Season

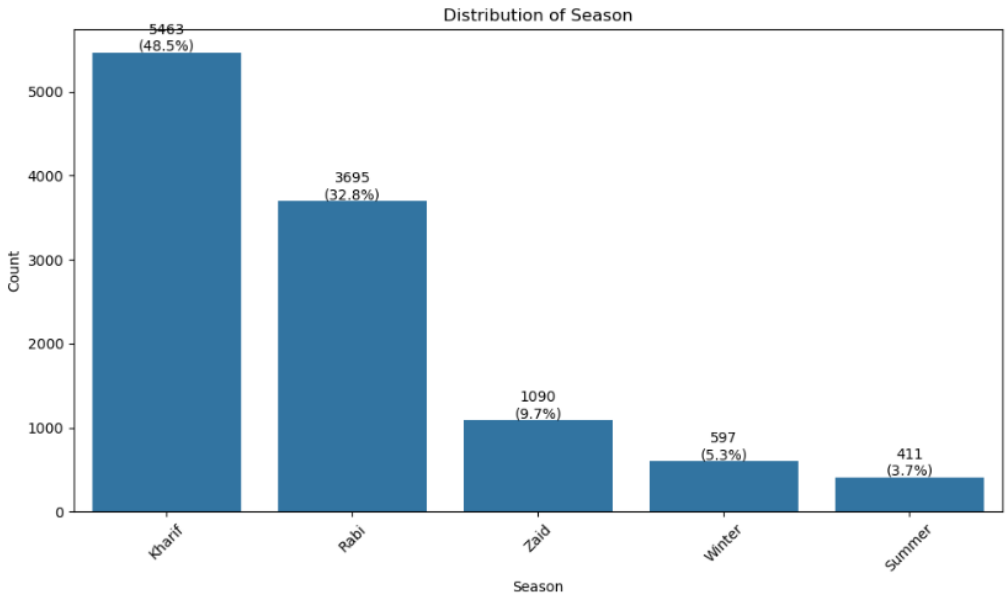

Fig 3.3 Distribution of season

Key Insights:

As shown in the above figure 3.2, the dataset is dominated by the Kharif season (48.5%)**,** followed by Other seasons like Zaid, Winter, and Summer have much lower representation, indicating a focus on major cropping seasons.

3.4.3 Distribution of SoilType

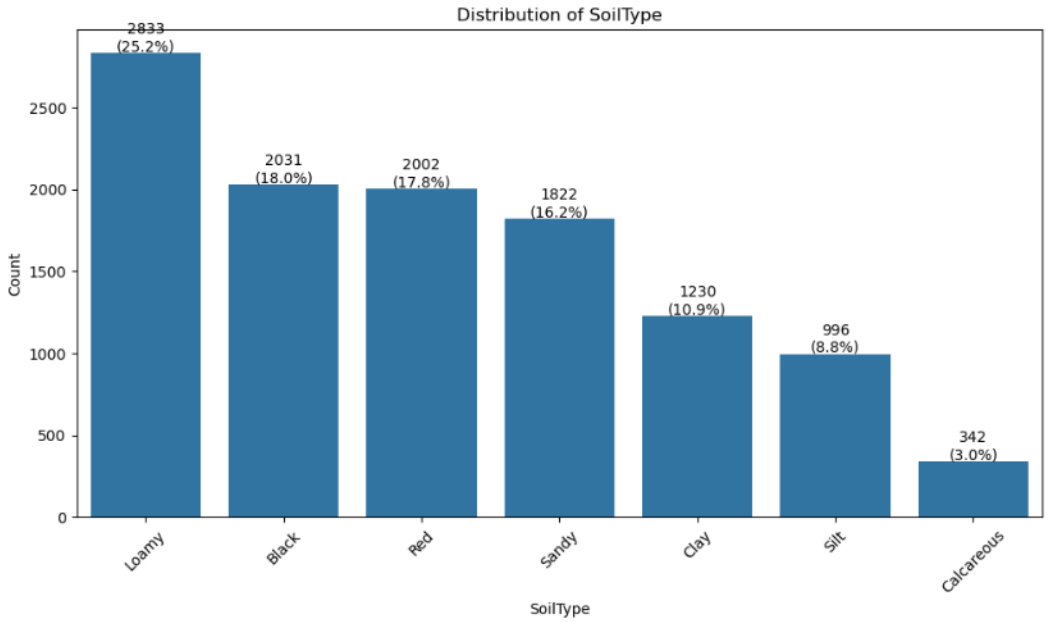

Fig 3.4 Distribution of soiltype

Key Insights:

As shown in the above figure 3.3, Loamy soil (25.2%) is the most common, followed by Black, Red, and Sandy soils. Other types like Clay, Silt, and Calcareous are less represented, showing moderate diversity in soil conditions.

3.4.4 Correlation Heatmap of Numerical features

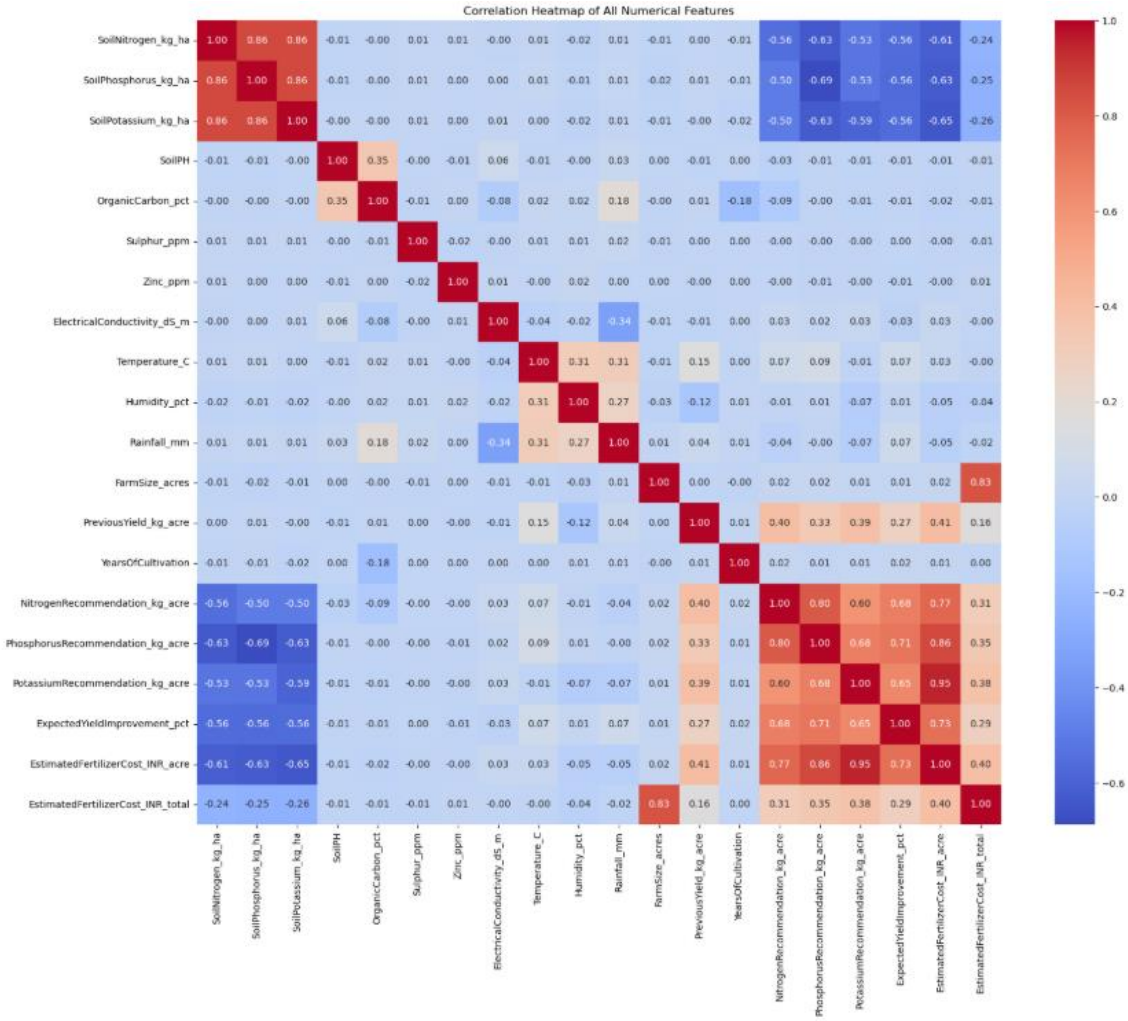

Fig 3.5 Correlation heatmap

Key Insights:

As shown in the above figure 3.4, a correlation heatmap was generated to examine the relationships between numerical features in the dataset. This visualization helps in identifying the strength and direction of relationships between variables, which is essential for feature selection and understanding data patterns.

From the heatmap, it is observed that soil nutrient features such as Nitrogen, Phosphorus, and Potassium show strong positive correlations among themselves, indicating that these nutrients often vary together. Additionally, the target variables - fertilizer recommendations for N, P, and K - also exhibit strong positive correlations with each other, suggesting interdependence in nutrient requirements.

Environmental factors such as temperature, humidity, and rainfall show moderate correlations with certain features, indicating their influence on soil conditions and fertilizer needs. However, most other features display weak or near-zero correlations, suggesting minimal direct linear relationships.

Some features, such as estimated fertilizer cost and expected yield improvement, show moderate to strong correlations with the recommendation variables, indicating their dependency on fertilizer quantities.

Overall, the heatmap reveals that while a few variables are strongly correlated, many features have low correlation, highlighting the complexity of the dataset. This justifies the use of advanced machine learning models capable of capturing non-linear relationships rather than relying solely on linear assumptions.

CHAPTER 4

DATA PREPARATION

The data preparation phase is a crucial step in the machine learning pipeline. It ensures that the dataset is clean, consistent, and properly structured for effective model training and evaluation. In this project, various preprocessing techniques were applied to transform raw agricultural and soil data into a suitable format for predicting fertilizer requirements.

4.1 DATA PREPROCESSING

Data preprocessing plays an important role in building an accurate and reliable machine learning model. The dataset used in this project contains over 11,000 records with multiple agricultural, soil, and environmental attributes**.** To ensure data quality and improve model performance, several preprocessing steps were carried out before model training.

4.1.1 Handling missing values

- Checked for missing/null values across all dataset attributes.
- As shown in the figure 4.1, missing values were handled based on the nature of the features:
- Numerical features such as Electrical Conductivity and Humidity were imputed using the median value of their respective columns to reduce the impact of outliers.
- Median imputation was preferred over mean to maintain robustness and avoid skewing the data distribution.
- This ensured that the dataset remained complete and suitable for machine learning model training without losing important information.

```
df['ElectricalConductivity_dS_m'] = df['ElectricalConductivity_dS_m'].fillna(df['ElectricalConductivity_dS_m'].median())
df['Humidity_pct'] = df['Humidity_pct'].fillna(df['Humidity_pct'].median())
```

Fig 4.1 Handling missing values

4.1.2 Encoding categorical variables

- As shown in the figure 4.2, categorical features were identified such as State, Crop, Season, Soil Type, and Irrigation Type**.**
- Applied One-Hot Encoding using `pd.get_dummies()` to convert categorical variables into numerical format.

- Used the parameter drop_first = True to avoid the dummy variable trap and reduce multicollinearity.
- This transformation enabled machine learning models to effectively interpret categorical data.

```
cat_cols = ['State','Crop', 'Season', 'SoilType', 'IrrigationType']
df= pd.get_dummies(df, columns=cat_cols, drop_first=True)
print(df.head())
```

Fig 4.2 Encoding categorical variables

4.1.3 Feature selection and Data splitting

- As shown in the figure 4.3, the target variables were identified as Nitrogen, Phosphorus, and Potassium recommendations**.**
- Removed irrelevant and non-informative columns such as FarmID, District, and cost-related features to avoid unnecessary influence on the model.
- Defined input features **(**X**)** and target variables (**y**) separately for model training.
- Split the dataset into training and testing sets using an 80:20 ratio.
- Used a fixed random state to ensure reproducibility of results.

```
target_cols = ['NitrogenRecommendation_kg_acre', 'PhosphorusRecommendation_kg_acre', 'PotassiumRecommendation_kg_acre']
exclude = target_cols + ['FarmID','District','EstimatedFertilizerCost_INR_acre', 'EstimatedFertilizerCost_INR_total', 'ExpectedYieldImprovement_pct']
X = df.drop(columns=exclude)
y = df[target_cols]
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Fig 4.3 Train-test split

4.1.4 Feature scaling

- As shown in the figure 4.4, Applied feature scaling using StandardScaler to normalize numerical features.
- The scaler was fitted on the training data and then used to transform both training and testing datasets.
- This ensured that all features have a standardized range with mean 0 and standard deviation 1.
- Scaling helps improve the performance and stability of machine learning models, especially those sensitive to feature magnitudes.

```python
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Fig 4.4 feature scaling

CHAPTER 5

MODEL BUILDING

5.1 MODEL BUILDING

Data modeling plays a crucial role in data analytics as it focuses on transforming prepared data into meaningful insights through the use of statistical and machine learning techniques. The primary objective of modeling is to develop accurate and reliable models that can identify patterns in data and make predictions based on those patterns.

The modeling process involves using the preprocessed dataset to train machine learning algorithms, which learn relationships between input features and target variables. These trained models can then be used to generate predictions and support decision-making. In this project, modeling is used to predict optimal fertilizer requirements based on soil, environmental, and crop-related parameters.

5.1.2 Key Benefits of Data Modeling in This Project

- Improved Accuracy:
  Machine learning models provide more accurate predictions of fertilizer requirements compared to traditional methods. By learning from historical data, the models can estimate precise values for Nitrogen, Phosphorus, and Potassium.
- Better Insights:
  The models help in understanding the relationship between soil nutrients, environmental factors, and fertilizer needs. This enables deeper insights into how different parameters influence crop nutrition.
- Optimization:
  The system helps in optimizing fertilizer usage by recommending the right quantity of nutrients. This can reduce overuse or underuse of fertilizers, leading to better crop productivity and cost efficiency.
- Practical Decision Support:
  The developed model assists users in making informed agricultural decisions by providing data-driven fertilizer recommendations based on specific input conditions.

As shown in the figure 5.1, Models used in this project are:

MODEL	IMPORTANCE
Linear regression	Serves as a baseline model to understand linear relationships between features and target variables.
Random forest regressor	Improves prediction accuracy by combining multiple decision trees and handling non-linear relationships effectively.
Gradient boosting regressor	Provides high accuracy by sequentially correcting errors of previous models; best performing model in this project.

Fig 5.1 Models used

The best model is selected based on evaluation metrics such as R^2 score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). These metrics indicate how accurately the model predicts the fertilizer requirements. A higher R^2 score and lower error values represent better model performance.

In addition to accuracy, computational efficiency is also considered. Some models may provide highly accurate predictions but require more time and resources for training and prediction. Therefore, both performance and efficiency are taken into account to select a model that is not only accurate but also practical for real-world agricultural applications.

```
models = {
    'Linear Regression': MultiOutputRegressor(LinearRegression()),
    'Random Forest': MultiOutputRegressor(RandomForestRegressor(n_estimators=100, random_state=42)),
    'Gradient Boosting': MultiOutputRegressor(GradientBoostingRegressor(random_state=42))
}
```

Fig 5.2 Defining models

```python
results = []

for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    r2 = r2_score(y_test, y_pred, multioutput='raw_values')
    avg_r2 = np.mean(r2)

    results.append({
        'Model': name,
        'R2_N': r2[0],
        'R2_P': r2[1],
        'R2_K': r2[2],
        'Average_R2': avg_r2
    })

results_df = pd.DataFrame(results)
print("Model Performance Results:")
print(results_df)
```

Fig 5.3 Calculating R^2 scores

```python
from sklearn.metrics import mean_absolute_error, mean_squared_error
results = []
print(f"{'Model Name':<25} | {'MAE':<8} | {'RMSE':<8}")
print("-" * 60)

for name, model in models.items():
    y_pred = model.predict(X_test)
    mae = mean_absolute_error(y_test, y_pred)
    rmse = np.sqrt(mean_squared_error(y_test, y_pred))
    r2 = r2_score(y_test, y_pred)
    results.append({
        'Model': name,
        'MAE': mae,
        'RMSE': rmse,
        'R2': r2
    })

    print(f"{name:<25} | {mae:<8.2f} | {rmse:<8.2f}")


df_results = pd.DataFrame(results).sort_values(by='MAE')
```

Fig 5.4 Calculating MAE & RMSE

5.2 LINEAR REGRESSION

Linear Regression is one of the most fundamental and widely used supervised learning algorithms for regression tasks. It models the relationship between independent variables (features) and dependent variables (targets) by fitting a linear equation to the observed data. The algorithm attempts to minimize the difference between predicted and actual values using methods such as least squares optimization.

In the context of this project, Linear Regression serves as a baseline model to understand the basic relationship between soil parameters and fertilizer requirements. It assumes that the relationship between inputs and outputs is linear, which may not always hold true in real-world agricultural data.

Advantages:

- Simple and easy to implement
- Computationally efficient and fast to train
- Provides easily interpretable results
- Useful as a benchmark model

Limitations:

- Assumes linearity between variables
- Unable to capture complex, non-linear relationships

- Sensitive to outliers and multicollinearity

Application in this project:
Linear Regression was used as an initial model to establish a baseline performance against which more complex models were compared.

5.3 RANDOM FOREST REGRESSOR

Random Forest Regressor is an ensemble learning technique that combines multiple decision trees to improve predictive performance. It uses a method called bagging (bootstrap aggregating), where multiple subsets of the dataset are created, and a separate decision tree is trained on each subset. The final prediction is obtained by averaging the outputs of all trees.

This approach significantly reduces overfitting and increases model robustness. Random Forest is particularly effective for structured datasets with multiple features and interactions.

Advantages:

- Reduces overfitting compared to a single decision tree
- Provides high accuracy and robustness
- Handles large datasets and high-dimensional data effectively
- Can estimate feature importance

Limitations:

- Less interpretable compared to individual trees
- Requires more memory and computational power
- Slower compared to simpler models

Application in this project:
Random Forest was implemented to improve generalization and achieve better predictive accuracy by leveraging multiple decision trees.

5.4 GRADIENT BOOSTING REGRESSOR

Gradient Boosting Regressor is an advanced ensemble learning technique that builds models sequentially. Unlike Random Forest, where trees are built independently, Gradient Boosting constructs each new tree to correct the errors made by the previous ones. It optimizes a loss function using gradient descent, making it highly effective for complex prediction tasks. This method is particularly powerful in capturing intricate patterns and relationships within the data. It balances bias and variance effectively, leading to improved model performance. Advantages:

High predictive accuracy

- Effectively captures complex and non-linear relationships
- Reduces both bias and variance
- Flexible and adaptable to different loss functions

Limitations:

- Computationally intensive and slower to train
- Sensitive to hyperparameter tuning
- Can overfit if not properly regularized

Application in this project:

Gradient Boosting Regressor achieved the best performance among all models based on evaluation metrics and was selected as the final model for fertilizer prediction.

CHAPTER 6

MODEL EVALUATION

6.1 MODEL EVALUATION

Model evaluation is a crucial step in the machine learning pipeline, as it determines how effectively a model performs in predicting the desired outcomes. In the context of this project, which focuses on predicting fertilizer requirements based on soil and environmental parameters, evaluation involves assessing how accurately and reliably the models estimate the target values.

Since this is a regression-based problem, the evaluation process emphasizes measuring the difference between predicted and actual values using appropriate statistical metrics. Proper evaluation ensures that the selected model is not only accurate but also capable of generalizing well to unseen data.

Models used:

- Linear regression
- Random forest regressor
- Gradient boosting regressor

6.1.1 Importance of Model Evaluation

Model evaluation serves several essential purposes in machine learning:

1. Selection of Optimal Model

Evaluation enables the identification of the most suitable model among multiple algorithms. By comparing models using metrics such as R^2 score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE), the best-performing model for fertilizer prediction can be selected.

2. Assessment of Model Performance

It provides insights into how well the model performs on unseen data. A model may perform well on training data but fail to generalize to new data, indicating overfitting. Evaluation helps in detecting such issues.

3. Comparative Analysis of Models

Different machine learning models exhibit varying strengths and limitations. Evaluation facilitates a systematic comparison of models such as Linear Regression, Random Forest, Gradient Boosting, and XGBoost, enabling informed decision-making.

4. Support for Decision Making

Evaluation metrics guide decisions related to model refinement, tuning, and deployment. If a model does not achieve the desired performance, further improvements such as hyperparameter tuning or feature engineering can be applied.

5. Practical Relevance and Impact

Effective evaluation ensures that the model can be applied reliably in real-world scenarios. Accurate fertilizer predictions can assist in optimizing resource usage, reducing costs, and improving agricultural productivity.

6.2 PERFORMANCE MEASUREMENT METRICS

For this regression-based problem, the following evaluation metrics were used:

1. R^2 Score (Coefficient of Determination)

The R^2 score measures the proportion of variance in the dependent variable that can be explained by the independent variables. It indicates how well the model fits the data.

- Value ranges between 0 and 1
- Higher values indicate better model performance

2. Mean Absolute Error (MAE)

MAE represents the average absolute difference between predicted and actual values. It provides a clear measure of prediction accuracy.

- Lower values indicate better performance
- Less sensitive to outliers

3. Root Mean Squared Error (RMSE)

RMSE calculates the square root of the average squared differences between predicted and actual values. It penalizes larger errors more significantly.

- Lower values indicate better model accuracy
- More sensitive to large deviations

Model evaluation is a fundamental component of this project, ensuring that the developed models are accurate, reliable, and suitable for practical implementation. By systematically analyzing performance using appropriate regression metrics, the most effective model was identified.

The evaluation process not only aids in selecting the best model but also provides deeper insights into model behavior, enabling continuous improvement and ensuring meaningful predictions for fertilizer recommendation.

6.3 MODEL PERFORMANCE ANALYSIS

Model Performance Results:

	Model	R2_N	R2_P	R2_K	Average_R2
0	Linear Regression	0.745751	0.715308	0.785583	0.748881
1	Random Forest	0.879731	0.829948	0.893322	0.867667
2	Gradient Boosting	0.887096	0.849612	0.895642	0.877450

Fig 6.1 R^2 score results

Model Name	MAE	RMSE
Linear Regression	5.69	7.83
Random Forest	3.65	5.54
Gradient Boosting	3.58	5.36

Fig 6.2 MAE & RMSE results

The performance of different machine learning models was evaluated using multiple metrics, including R^2 score**,** Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE)**.** These metrics provide a comprehensive understanding of model accuracy and error levels in predicting fertilizer requirements for Nitrogen (N), Phosphorus (P), and Potassium (K).

As shown in the figure 6.1 and figure 6.2,

- Gradient Boosting emerged as the best-performing model, achieving the highest average R^2 score of approximately 0.877 along with the lowest error values (MAE = 3.58 and RMSE = 5.36). This indicates that the model provides highly accurate predictions with minimal deviation from actual values.
- The Random Forest also demonstrated strong performance. It achieved an average R^2 of 0.868 with MAE = 3.65 and RMSE = 5.54. These results highlight the effectiveness of ensemble-based approaches in handling complex and non-linear data.
- The Linear Regression model showed moderate performance with an average R^2 of 0.749, along with higher error values (MAE = 5.69**,** RMSE = 7.83), indicating that linear assumptions are insufficient to fully capture the relationships in the dataset.

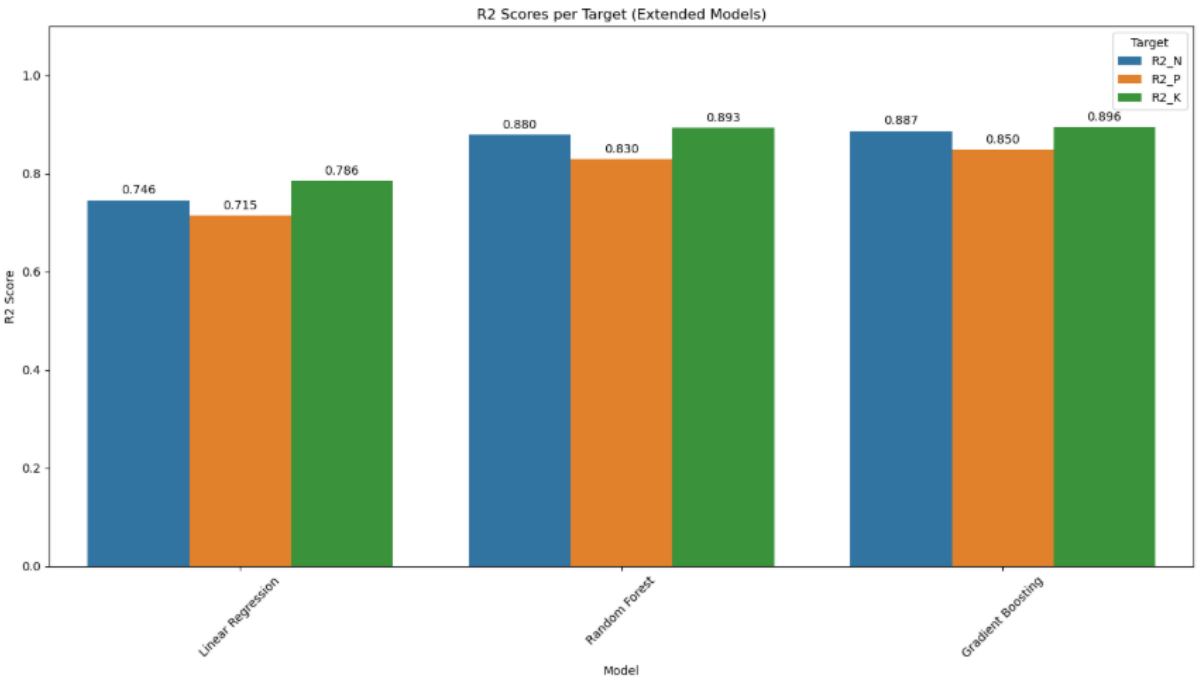

Fig 6.3 Barplot comparison of individual R^2 scores

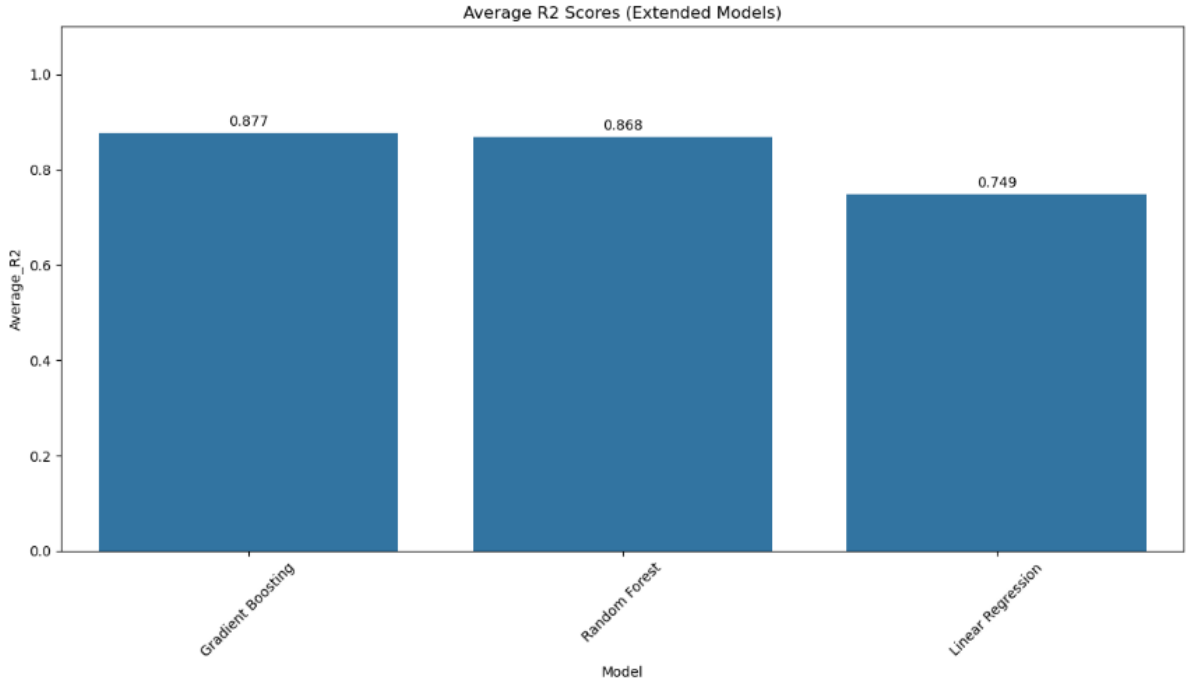

Fig 6.4 Barplot comparison of average R^2 scores

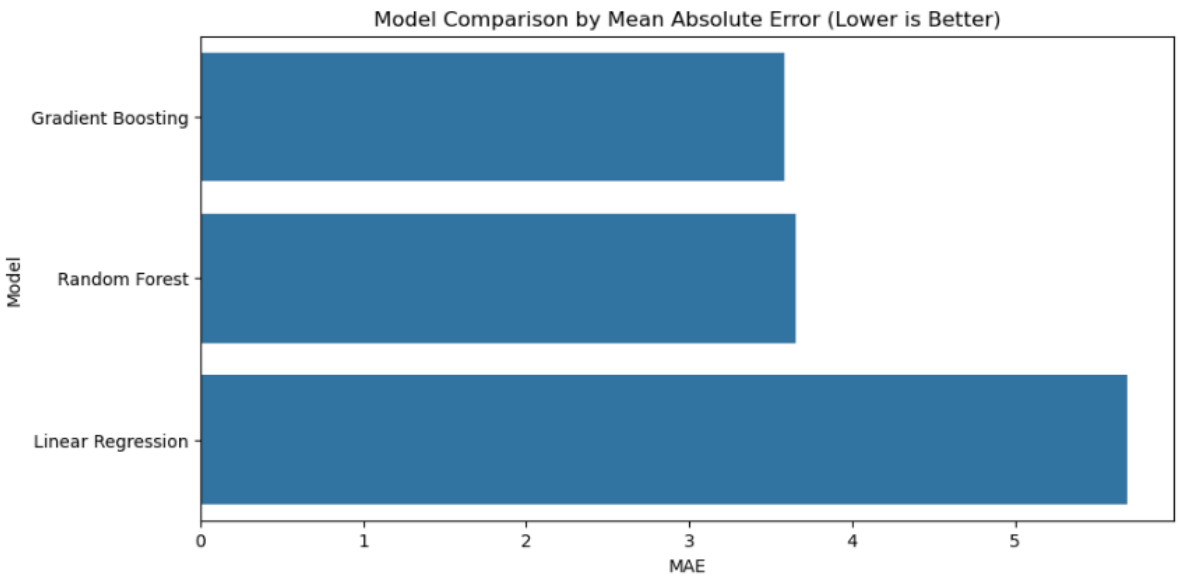

Fig 6.5 Barplot comparison of MAE

As shown in the figures 6.3, 6.4, 6.5, the evaluation results clearly indicate that ensemble learning methods, particularly Gradient Boosting, provide superior performance in terms of both accuracy and error minimization. Therefore, Gradient Boosting was selected as the final model for fertilizer prediction.

6.4 FEATURE IMPORTANCE

Feature importance analysis was conducted to identify the most influential variables in predicting fertilizer recommendations for Nitrogen (N), Phosphorus (P), and Potassium (K). This analysis helps in understanding which features contribute the most to the model's decision-making process.

As shown in the figure 6.6, it is evident that Soil Phosphorus is the most significant feature, followed by Previous Yield**,** Soil Potassium, and Soil Nitrogen. These features have the highest importance scores, indicating that existing soil nutrient levels and past yield play a crucial role in determining fertilizer requirements.

Crop-related features such as Groundnut, Sugarcane, Potato, and Soybean also show moderate importance, suggesting that fertilizer needs vary depending on the type of crop being cultivated.

Environmental and soil condition factors like Organic Carbon, Irrigation Type, and Soil pH contribute to the model but with relatively lower importance compared to primary nutrient features. This indicates that while these factors influence fertilizer recommendations, their impact is secondary.

Other variables, including state, season, and certain soil types, exhibit minimal importance, suggesting that they have a limited direct effect on fertilizer prediction in this dataset.

Overall, the analysis highlights that soil nutrient composition and historical yield data are the most critical factors influencing fertilizer recommendations. This insight aligns with agricultural principles and validates the effectiveness of the model in capturing real-world relationships.

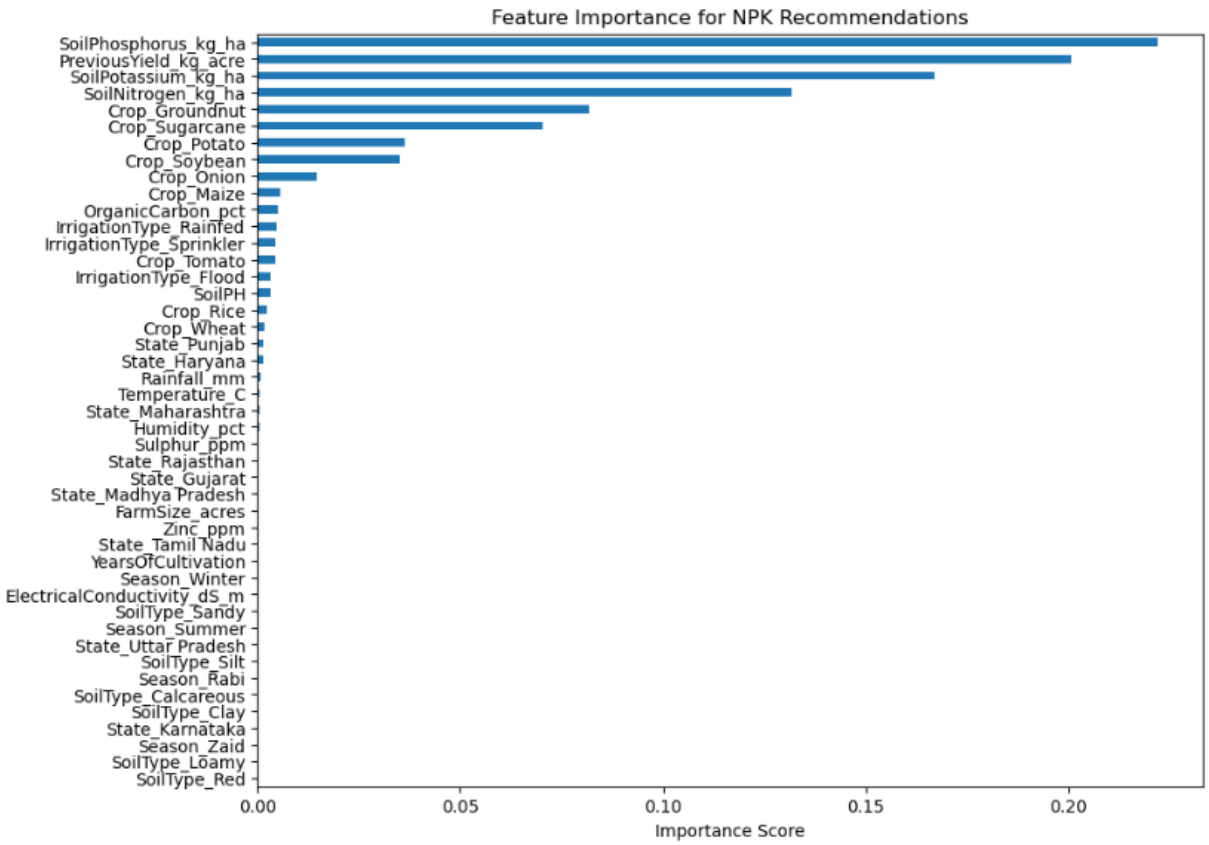

Fig 6.6 Feature importance graph

6.5 ACTUAL VS PREDICTED

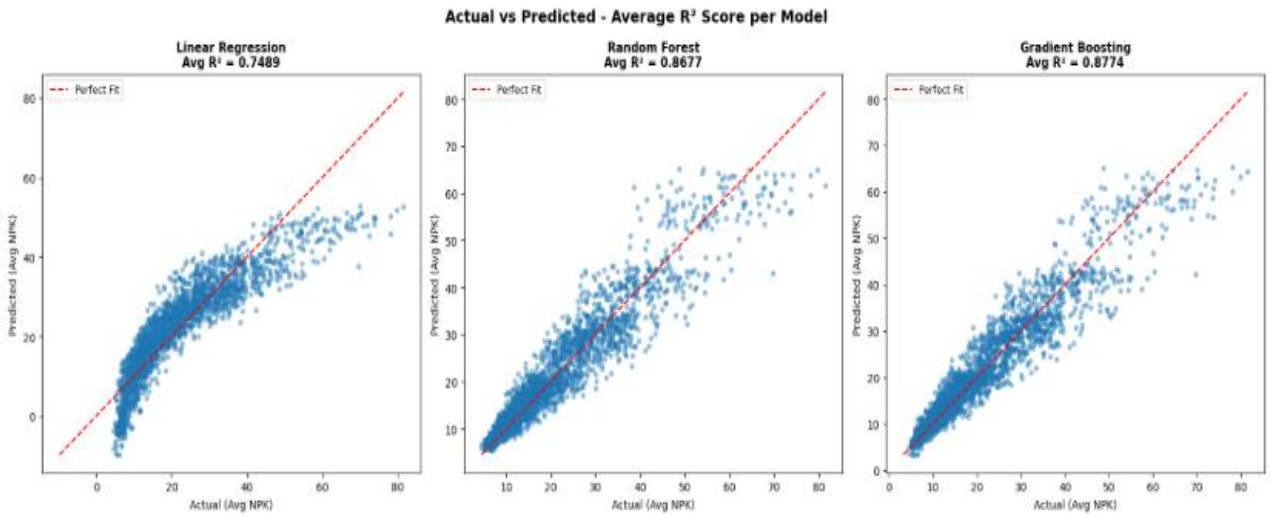

Fig 6.7 Actual vs Predicted

The actual vs predicted analysis was conducted to evaluate the performance of the three selected models — Linear Regression, Random Forest, and Gradient Boosting — in predicting the NPK (Nitrogen, Phosphorus, and Potassium) fertilizer recommendations.

A scatter plot was generated for each model, where the X-axis represents the actual values and the Y-axis represents the predicted values. A red dashed reference line indicating perfect prediction (where actual = predicted) was overlaid on each plot. The proximity of the scatter points to this reference line serves as a visual indicator of model accuracy.

Interpretation of Results

As shown in the figure 6.7, the scatter plot for Linear Regression shows a moderate spread of points around the perfect fit line. While a general linear trend is visible, several data points deviate significantly from the reference line, indicating limited predictive accuracy. This is reflected in its Average R^2 score of 0.7489, which, while acceptable, is the lowest among the three models evaluated.

The Random Forest model demonstrates a noticeably tighter clustering of points around the perfect fit line compared to Linear Regression. The predictions are more consistent across the

range of actual values, with fewer extreme deviations. This improvement in visual accuracy is supported by its Average R^2 score of 0.8677, indicating that the model captures a significantly higher proportion of variance in the target variables.

The Gradient Boosting model produces the tightest and most uniform distribution of scatter points along the perfect fit line among all three models. The predictions closely follow the actual values across the entire range, with minimal over-prediction or under-prediction. This is confirmed by the highest Average R^2 score of 0.8775, establishing Gradient Boosting as the best performing model in this study.

CHAPTER 7

DEPLOYMENT

Overview of Deployment

The final phase of the CRISP-DM methodology involves deploying the developed model into a usable application. In this project, the trained Gradient Boosting model was deployed as an interactive web application using Gradio, an open-source Python library that enables rapid prototyping of machine learning interfaces. The deployed application is titled NPK Fertilizer Recommendation System and is accessible via a shareable web link without requiring any installation on the user's end.

7.1 TOOLS AND TECHNOLOGIES USED

The deployment was built using the following tools and technologies:

- Gradio (gr.Blocks) ─ for building the interactive web interface with a soft green theme using the DM Sans font, suited to the agricultural context of the project.
- Pickle (.pkl) — the trained Gradient Boosting model was serialized and saved as npk_model.pkl for instant loading without retraining.
- JSON (.json) — metadata including feature column names, dataset medians, R^2 scores, category lists and number of training records were stored in npk_meta.json for dynamic use within the application.
- Pandas and NumPy — used for constructing the input feature row and processing predictions at runtime.

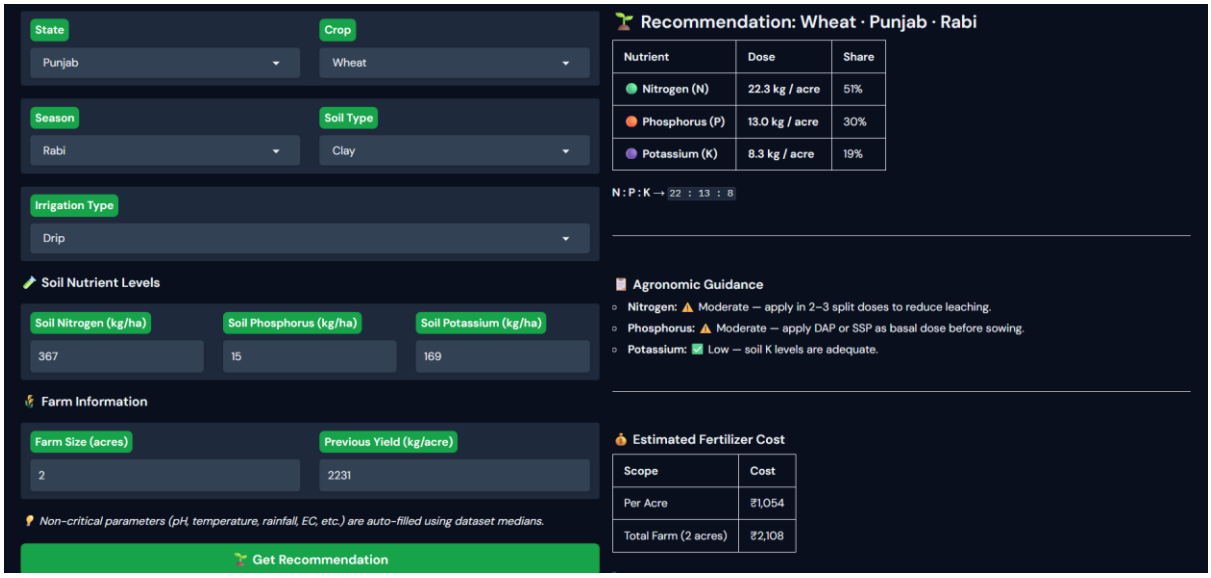

Fig 7.1 User interface of gradio app

7.2 APPLICATION OF ARCHITECTURE

As shown in the figure 7.1, the application follows a three-component architecture:

Component 1 — Model and Metadata Loading: Upon launch, the application loads the pre-trained Gradient Boosting model from `npk_model.pkl` and the associated metadata from `npk_meta.json`. This includes feature column names, dataset medians for auto-filling non-critical parameters, R^2 scores, and category lists for all dropdown inputs. This design ensures the application starts instantly without any retraining overhead.

Component 2 — Prediction Function: The core prediction function accepts ten user inputs — State, Crop, Season, Soil Type, Irrigation Type, Soil Nitrogen, Soil Phosphorus, Soil Potassium, Farm Size, and Previous Yield. The function constructs a complete feature row by filling the five primary numeric inputs from the user and auto-filling all remaining non-critical parameters such as pH, temperature, rainfall, and electrical conductivity using dataset medians. Categorical variables such as State, Crop, Season, Soil Type, and Irrigation Type are one-hot encoded at runtime to match the format used during model training.

Component 3 — Output Generation: Once the model generates predictions for N, P, and K values in kg/acre, the application performs the following additional computations:

- Calculates the N:P:K ratio using the GCD method for simplified representation
- Computes the percentage share of each nutrient
- Estimates the fertilizer cost per acre and for the total farm using approximate market rates (Urea ₹7/kg-N, DAP ₹55/kg-P, MOP ₹22/kg-K)
- Provides agronomic guidance for each nutrient based on the predicted dose range

7.3 USER INTERFACE DESCRIPTION

The application interface is divided into two panels:

Left Panel — Input Section**:** The left panel is organized into three subsections. The Farm & Crop Details section contains dropdown menus for State, Crop, Season, Soil Type, and Irrigation Type. The Soil Nutrient Levels section accepts current soil nitrogen, phosphorus, and potassium readings in kg/ha. The Farm Information section takes farm size in acres and previous yield in kg/acre. A primary action button labeled "Get Recommendation" triggers the prediction.

Right Panel — Results Section: The results are displayed as a formatted markdown output containing a nutrient recommendation table with doses and percentage shares, the simplified N:P:K ratio, agronomic guidance for each nutrient, estimated fertilizer cost breakdown, and the model's R^2 performance metrics.

The application also includes a Quick Examples section at the bottom with five pre-filled farm scenarios covering diverse states and crops including Wheat (Punjab), Cotton (Maharashtra), Rice (Tamil Nadu), Sugarcane (Uttar Pradesh), and Tomato (Karnataka). Users can click any example row to auto-fill all inputs and instantly view results.

7.4 LIMITATIONS OF PROTOTYPE

While the deployed application serves as a functional prototype, the following limitations are acknowledged:

- The shareable Gradio link is valid for 72 hours per session and requires the host machine to remain active, making it unsuitable for permanent production deployment.

- Non-critical parameters such as soil pH, temperature, and rainfall are auto-filled using dataset medians, which may reduce prediction accuracy for farms with significantly different environmental conditions.
- The cost estimates are based on approximate market rates and may not reflect current or region-specific fertilizer prices.
- The model is trained on data from 10 Indian states and may not generalize accurately to regions outside this scope.

CONCLUSION

This project focused on developing a data-driven fertilizer recommendation system to predict optimal NPK doses for Indian farming conditions, addressing the issue of blanket fertilizer application and its negative impact on soil health and crop productivity. The CRISP-DM framework structured the workflow, ensuring a systematic approach from problem understanding to deployment.The dataset included 11,256 farm records from ten Indian states, covering variables such as soil nutrients, crop types, irrigation methods, and seasonal conditions. Exploratory analysis highlighted that fertilizer requirements depend on the interaction of multiple factors rather than isolated variables, emphasizing the need for models capable of capturing complex relationships.

Three models were implemented: Linear Regression as a baseline, and Random Forest and Gradient Boosting as advanced ensemble methods. Among these, Gradient Boosting performed best, achieving an average R^2 score of 0.8775 across nitrogen, phosphorus, and potassium predictions. Visual comparisons of actual versus predicted values confirmed its superior accuracy and lack of systematic bias. During preprocessing, categorical encoding, feature handling, and median imputation were carefully applied to maintain data integrity and model reliability. These steps contributed to strong generalization performance on unseen data.

The project concluded with the deployment of a Gradio-based web application, allowing users to input farm-specific details and receive fertilizer recommendations in kg/acre, along with N:P:K ratios, agronomic guidance, and estimated costs. This makes the system practical, interpretable, and useful for real-world agricultural decision-making.

The system offers benefits such as reduced fertilizer overuse, cost savings, and improved soil health, supporting sustainable farming practices. However, its effectiveness depends on the quality of training data, and cost estimates may vary due to regional price differences.

Overall, the project demonstrates the potential of machine learning in precision agriculture, showing how data-driven solutions can improve efficiency, reduce waste, and support informed decision-making in modern farming.

FUTURE SCOPE

Future enhancements of the system can focus on integrating real-time data sources such as IoT-based soil sensors and weather APIs to deliver dynamic, location-specific recommendations. The incorporation of advanced machine learning and deep learning models can further improve prediction accuracy and overall system performance. Additionally, expanding the platform to include crop yield prediction and disease detection would make it a more comprehensive agricultural support tool. Developing a mobile application would enhance accessibility, especially for farmers in remote areas, while the inclusion of regional agricultural guidelines and government recommendations would ensure the system remains adaptable to diverse farming practices.

BIBILIOGRAPHY

1. **https://panapps.co/**
2. **https://www.geeksforgeeks.org/** - Machine learning tutorials.
3. Gradio Documentation. Available at: https://gradio.app
4. Géron, A. (2019). *Hands-On Machine Learning with Scikit-Learn*.
5. You tube tutorials.

ANNEXURE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.multioutput import MultiOutputRegressor
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor, AdaBoostRegressor, ExtraTreesRegressor
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score
from sklearn.tree import DecisionTreeRegressor
import xgboost as xgb
```

```
df=pd.read_csv("C:/Users/ANAGHA BABU/Desktop/PROJECT/fertilizer_utilization_dataset_v3_realistic.csv")
df
```

```
df.shape
```

```
(11256, 27)
```

```
df.dtypes
```

```
df.info()
```

```
df.describe()
```

```
print("\nDuplicate Rows:", df.duplicated().sum())
```

```
for col in df.columns:
    print(f"\nColumn: {col}")
    print(df[col].nunique())
```

```
df.isnull().sum()
```

```
plt.figure(figsize=(20, 15))
corr_matrix = df.corr(numeric_only=True)
sns.heatmap(corr_matrix, annot=True, fmt='.2f', cmap='coolwarm', square=True)
plt.title('Correlation Heatmap of All Numerical Features')
plt.show()
plt.close()
```

```
%matplotlib inline
import matplotlib.pyplot as plt
import seaborn as sns

cat_cols_for_eda = ['State', 'Crop', 'Season', 'SoilType', 'IrrigationType']

def plot_categorical_univariate(df, cat_cols):
    for col in cat_cols:
        fig = plt.figure(figsize=(10, 6))

        ax = sns.countplot(data=df, x=col, order=df[col].value_counts().index, color='#1F77B4')

        total = len(df[col])
        for p in ax.patches:
            height = p.get_height()
            percentage = '{:.1f}%'.format(100 * height / total)
            ax.annotate(f'{int(height)}\n({percentage})',
                        (p.get_x() + p.get_width() / 2., height),
                        ha='center', va='center', fontsize=10,
                        color='black', xytext=(0, 10),
                        textcoords='offset points')

        plt.title(f'Distribution of {col}')
        plt.ylabel('Count')
        plt.xticks(rotation=45)
        plt.tight_layout()
        plt.show()
        plt.close(fig)

plot_categorical_univariate(df, cat_cols_for_eda)
```

```
num_cols = df.select_dtypes(include=np.number).columns
```

```
def check_outliers_iqr(data):
    outlier_summary = {}

    for col in data.columns:
        Q1 = data[col].quantile(0.25)
        Q3 = data[col].quantile(0.75)
        IQR = Q3 - Q1

        lower = Q1 - 1.5 * IQR
        upper = Q3 + 1.5 * IQR

        outliers = data[(data[col] < lower) | (data[col] > upper)]
        outlier_summary[col] = len(outliers)

    return pd.DataFrame(outlier_summary.items(), columns=["Feature", "Outlier_Count"])

outlier_report = check_outliers_iqr(df[num_cols])
print(outlier_report)
```

```
%matplotlib inline
plot_data = results_df.melt(id_vars='Model', value_vars=['R2_N', 'R2_P', 'R2_K'], var_name='Target', value_name='R2 Score')

plt.figure(figsize=(14, 8))
sns.barplot(data=plot_data, x='Model', y='R2 Score', hue='Target')
plt.title('R2 Scores per Target (Extended Models)')
plt.ylim(0, 1.1)
plt.xticks(rotation=45)
for i in plt.gca().containers:
    plt.gca().bar_label(i, fmt='%.3f', padding=3)
plt.tight_layout()
plt.show()
```

```
plt.figure(figsize=(12, 7))
results_df_sorted = results_df.sort_values(by='Average_R2', ascending=False)
ax = sns.barplot(data=results_df_sorted, x='Model', y='Average_R2',color='#1F77B4' )
plt.title('Average R2 Scores (Extended Models)')
plt.ylim(0, 1.1)
plt.xticks(rotation=45)
for i in ax.containers:
    ax.bar_label(i, fmt='%.3f', padding=3)
plt.tight_layout()
plt.show()
```

```
best_model = models["Gradient Boosting"]
avg_importances = np.mean([est.feature_importances_ for est in best_model.estimators_], axis=0)
feat_importances = pd.Series(avg_importances, index=X.columns).sort_values(ascending=True)

plt.figure(figsize=(10, 8))
feat_importances.plot(kind='barh', color='#1F77B4')
plt.title('Feature Importance for NPK Recommendations')
plt.xlabel('Importance Score')
plt.show()
```

```
plt.figure(figsize=(10, 5))
sns.barplot(data=df_results, x='MAE', y='Model', color='#1F77B4' )
plt.title('Model Comparison by Mean Absolute Error (Lower is Better)')
plt.show()
```